

Khulna University of Engineering & Technology  
B. Sc. Engineering 4<sup>th</sup> Year 1<sup>st</sup> Term Regular Examination, 2023  
Department of Materials Science and Engineering  
MSE 4101  
Modern Iron and Steel Making

Time: 3 hours

Full Marks: 210

- N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figures in the right margin indicate full marks.

**Section A**

- Q1. (a) Show all the possible relations taking place mentioning the temperatures in different regions of a blast furnace with proper diagram. (12)  
(b) Explain how and where impurities (Si, Mn, S) are picked up in liquid iron in blast furnace. (11)  
(c) Apply your understanding of thermal expansion to justify the need for increasing the cross-sectional area of the furnace as the charge descends from the stockline to the mantle level. (12)
- Q2. (a) Analyze the significance of maintaining a solid burden in the shaft or stack zone of the blast furnace for the overall efficiency of the smelting process. (11)  
(b) How raceway is created? Explain the importance of inverted U-shaped zone in blast furnace. (11)  
(c) “The higher the iron content in the feed, the lower is the slag volume generated in BF, which automatically increases the productivity and reduces the coke rate.” – Justify the statement explaining the feed making process. (13)
- Q3. (a) Given a scenario where the slag in a steel making process has a high viscosity, propose adjustments to the temperature and slag composition to improve its fluidity. (15)  
(b) What are scaffold irregularities in a blast furnace, and how they form? Identify the factors contributing to scaffold formation and describe the specific process changes you would implement to restore normal operating conditions in furnace. (10)  
(c) Briefly describe the different techniques of thermal characterization of BF feedstock (lump ore, sinter, pellets). (10)
- Q4. (a) Discuss the electrothermal process of pig iron production and how it differs from traditional blast furnace methods. (07)  
(b) Compare and contrast HYL III and HYL IV process. (08)  
(c) What are the key differences between Sponge Iron Production in Rotary Kilns and the Midrex process? Based on these differences, which process would you recommend for implementation in Bangladesh, considering the local resources and industrial conditions? (20)

**Section B**

- Q5. (a) Explain the importance of pretreatment of hot metal prior to steel making in Basic Oxygen Furnace. (13)  
(b) What are the basic differences between a Basic Oxygen Furnace (BOF) and an Electric Arc Furnace (EAF)? Show neat sketch. (12)  
(c) Calculate the values of  $[H]$  when molten steel is degassed at 1850 K under (10)

vacuum. The pressure above is reduced in one case to 1 mm Hg and in other cases to 0.1 and 10 mm Hg. Steel contains C = 0.05%, Cr = 6%, Ti = 0.6%, Ni = 2%. Assume  $e_H^{Ni} = 0$ .

- Q6. (a) When and how was double slag practice of EAF steel making process adopted (10)  
instead of single slag practice?
- (b) Explain the thermodynamics of decarburization of chromium melt. (12)
- (c) Suppose you are hired as a metallurgist to produce very low C and S level (13)  
stainless steel. Between EAF and EAF+AOD which process you would suggest?  
Justify your suggestion.
- Q7. (a) With proper illustration, explain the stream degassing processes. (10)
- (b) Outline the main steps of degassing practices by ladle and stream. Which one is (15)  
better between ladle and stream degassing? Justify your answer.
- (c) A 0.1% C steel strip 50 mm wide, 5 mm thick was rolled in one pass to 3.5 mm at (10)  
1060°C when the homogeneous yield stress was 1.05 KN/mm<sup>2</sup>. The roll diameter  
was 340 mm. Find the magnitude of the rolling load, taking into account roll  
flattening, if the rolls were made of cast iron.
- Q8. (a) State the central problem in stainless steel making and explain how it can be (10)  
solved.
- (b) How can you differentiate between a pipe and blow-hole defect, and what are the (10)  
possible remedies for each?
- (c) A 300 mm wide strip of 25 mm thick is fed through a rolling mill with two (15)  
powered rolls each of radius = 250 mm. The work thickness is to be reduced to 22  
mm in one pass at a roll speed of 50 rev/min. The work material has a flow curve  
defined by  $K = 275$  MPa and  $n = 0.15$ , and the coefficient of friction between the  
rolls and the work is assumed to be 0.12. Determine if the friction is sufficient to  
permit the rolling operation to be accomplished. If so, calculate the roll force,  
torque and horse power. N.B. Avg. flow stress,  $\sigma_0 = k\epsilon^n / (n + 1)$ .

Khulna University of Engineering & Technology  
B. Sc. Engineering 4<sup>th</sup> Year 1<sup>st</sup> Term Regular Examination, 2023  
Department of Materials Science and Engineering  
MSE 4103

Engineering Alloys and Materials Selection

Time: 3 hours

Full Marks: 210

- N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figures in the right margin indicate full marks.

**Section A**

- Q1. (a) Name a type of maraging steel which is weldable and shows good machinability in annealed condition. Also explain the heat treatment process of the suggested steel with schematic diagram. (14)  
(b) List the carbide phases present in stainless steels. Explain in reference to the phase diagram why SS containing 18% Cr is regarded as non-heat treatable whereas SS containing 12% Cr is heat-treatable. (10)  
(c) “Tool steels should always be at least double tempered”. Explain the quotation. (11)
- Q2. (a) A typical off-shore pipeline is made of HSLA steels. Determine the reasons for the unusual properties of HSLA mentioning a typical composition of the HSLA. (13)  
(b) Explain Sub-Zero treatment. (10)  
(c) Compare the mechanical properties of Cu-Zn and Cu-Sn system and explain the underlying differences between them. (12)
- Q3. (a) “Mg-Mn alloy can be strengthened by solid solution strengthening whereas Mg-Al-Zn alloy can be strengthened by other means”. Do you agree with the statement? Explain the reasons with reference to phase diagram of the alloys. (12)  
(b) Figure Q3(b) shows two microstructures of the alloy A356. Explain the difference in the two microstructures and suggest a way to convert the microstructure from one another. (10)

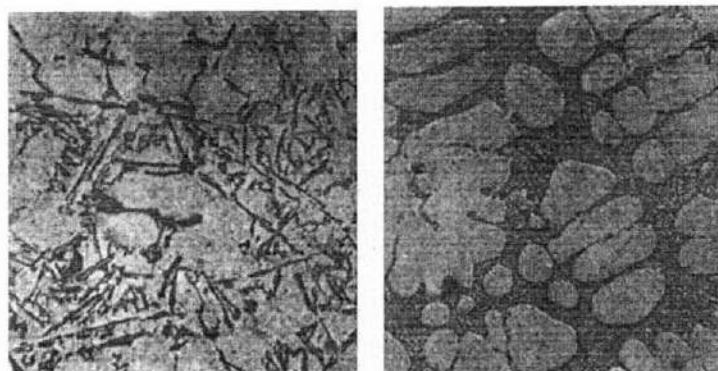


Figure for Q3(b)

- (c) Suggest an alloy for applications in the biomedical sector. Draw schematic diagram and explain the basic principles of heat treatment of the alloy. (13)
- Q4. (a) Write short note on “Nickel Silver”. (10)  
(b) Discuss briefly about sensitization and explain the reason behind sensitization. List the remedies of sensitization and name an alloy where sensitization is often encountered. (10)  
(c) Interpret the shape recovery of NiTiNOL and mention some applications of NiTiNOL. (15)

## Section B

- Q5. (a) "Market pull, technology push" Justify the statement. (10)  
 (b) Draw a design flowchart to show the influence of design tools and materials selection. (10)  
 (c) An industry needs to design an eco-friendly packaging solution. Identify the primary considerations for materials selection with respect to environmental impact. Suggest a material and explain how it meets the criteria for sustainability, recyclability, and cost-effectiveness. (15)
- Q6. (a) A structural component made of aluminium alloy is subjected to tensile load of 50,000 N. The cross-sectional area of the component is  $25 \text{ mm}^2$ . Determine whether it is safe to use if the yield strength of the alloy is 200 MPa. (10)  
 (b) Calculate the material index for light, stiff beam. (10)  
 (c) In case of multiple constraint problem of materials selection: Explain and compare the analytical method and graphical method. (15)
- Q7. (a) Derive the expression for the shape efficiency factor  $\Phi_B^e$  for stiffness-limited design for a beam loaded in bending with each of the two sections listed below for the Figure Q7(a). (10)  
 (i) A channel section of thickness  $t$ , overall flange width  $4t$  and overall depth  $8t$ , bent about its major axis; and  
 (ii) A rectangular wall of thickness  $t$ , and height and width are  $8t$  and  $12t$ .

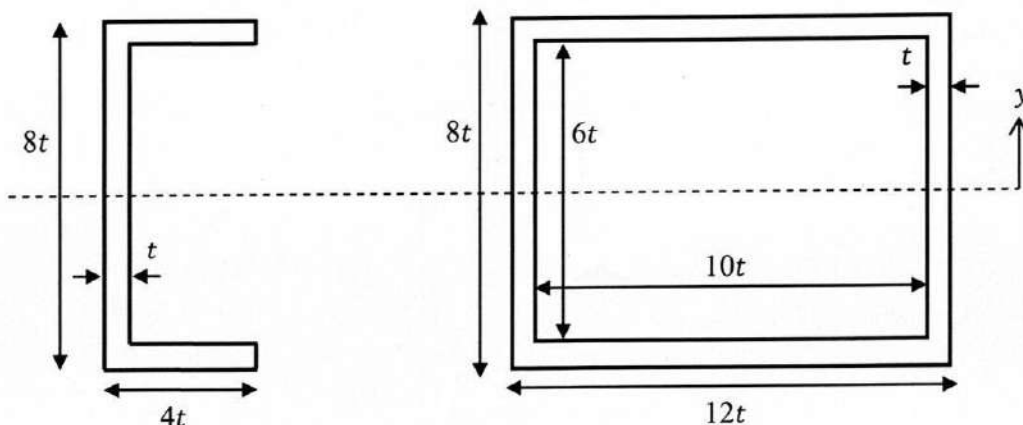


Figure for Q7(a)

- (b) The design requirement for selecting a connecting rod for high performance engine is given below. (25)

| The design requirements: Connecting Rods |                                                                                                                            |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Function                                 | : Connecting rod for reciprocating engine                                                                                  |
| Constraints                              | : Must not fail by high-cycle fatigue<br>Must not fail by elastic buckling stroke, and thus con-rod length $L$ , specified |
| Objective                                | : Minimize mass                                                                                                            |
| Free variable                            | : Cross Section A<br>Choice of Materials                                                                                   |

- (i) Select a material using min-max method. Follow Figure Q7(b) and Table Q7(b). Assume length  $L = 100 \text{ mm}$ , load  $F = 25 \text{ kN}$ , dimensionless shape constant  $\alpha = 0.4$  and  $\beta = 0.75$ .  
 (ii) Show the equations schematically for the mass of the con-rod as a function of  $L$ .

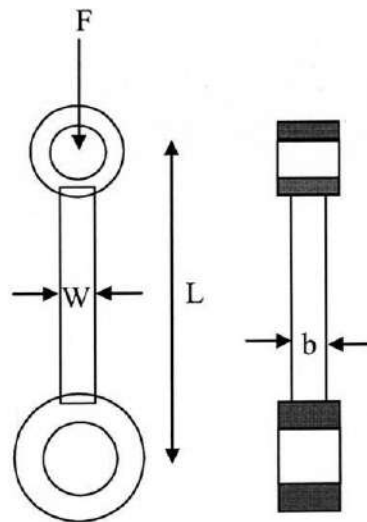


Figure for Q7(b)

Table for Q7(b): Selection of a material for the con-rod

| Material                | $\rho$ Kg/m <sup>3</sup> | E GPa | $\sigma_e$ MPa |
|-------------------------|--------------------------|-------|----------------|
| Nodular Cast Iron       | 7150                     | 178   | 250            |
| HSLA Steel 4140         | 7850                     | 210   | 590            |
| Al S355.0 Casting Alloy | 2700                     | 70    | 95             |
| Composite               | 2880                     | 110   | 230            |
| Titanium 6Al 4V         | 4400                     | 115   | 530            |

- Q8. (a) Explain the concept of penalty function. (11)
- (b) Among single chart, 4-quadrant charts and indices including SF charts, which one is better in your opinion? Justify your answer. (12)
- (c) Discuss percolation and its importance as a design tool for hybrid materials. (12)

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B. Sc. Engineering 4<sup>th</sup> Year 1<sup>st</sup> Term Regular Examination, 2023  
Department of Materials Science and Engineering  
MSE 4105

Polymer and Composite Materials

Time: 3 hours

Full Marks: 210

- N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figures in the right margin indicate full marks.

**Section A**

- Q1. (a) If the number average degree of polymerization,  $X_n$  for nylon is 100 and  $M_w = 18300$ . What is its polydispersity? Draw the molecular weight curve for this polymer. The structure of monomer is shown in Fig. 1(a). (13)

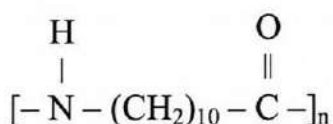


Fig. 1(a): The monomer of nylon.

- (b) Explain the following observation. When thalic acid reacts with glycerol, the reaction leads first to the formation of fairly soft soluble material, which on further heating yields a hard, insoluble and infusible material. If the same reaction is carried out with ethylene glycol instead of glycerol, the product remains soluble and fusible irrespective of the extent of reaction. (15)
- (c) What are polymer additives and why these additives are used in polymer processing? (07)
- Q2. (a) The following structure (Fig. 2(a)) represents the general formula for some aliphatic amines. What is the functionality of the corresponding amine in its reaction with diglycidyl ether of bisphenol A (DGEBA) if  $n = 1, 2, 3, 4$ ? (13)



Fig. 2(a): Aliphatic amines.

- (b) Briefly explain the main step of ring opening polymerization with proper examples. (10)
- (c) Deduce the kinetics of free radical polymerization and derive the relationship between the  $R_p$ ,  $[I]$  and  $[M]$ . (12)
- Q3. (a) The thermoplastic elastomers have replaced conventional thermoset elastomers for the electrical insulation of automotive components. Justify the statement with diagram. (10)
- (b) Explain the deformation of a semicrystalline polymer with suitable diagram when tensile load is applied to the polymer. (13)
- (c) Explain the vulcanization process with proper example and how does it affect the mechanical properties of polymer? (12)
- Q4. (a) Design and explain a fabrication process that is used to form a hollow polymeric pipe. (15)
- (b) Why liquid crystal polymer is used for the making of display of mobile phone (12)



instead of semicrystalline and amorphous polymer? Explain with suitable diagram.

- (c) How does the addition of stabilizer protect the polymer from degradation and why? (08)

## Section B

- Q5. (a) What are the matrix and dispersed phase in a composite? Briefly explain the functions of the matrix materials. (08)
- (b) You have created a composite system with a new kind of ballistic fiber. The fibers are continuous, aligned and brittle in nature, whereas the matrix is ductile. A load  $F$  is applied along the transverse direction. For the fiber volume of  $V$ , derive the expression of the modulus of elasticity for your composite system. (12)
- (c) For the new composite system in Q5(b), suppose you have measured the stress-strain behaviour of the fiber, matrix and composite, and obtained the Fig. 5(a) below. (15)
- (i) Explain how the fiber, matrix and composite will behave in stage I and stage II region.
- (ii) You find that new composite shows a failure when strain  $> \epsilon_f^*$ . Will this be a catastrophic or non-catastrophic failure? Provide reasoning for your answer.

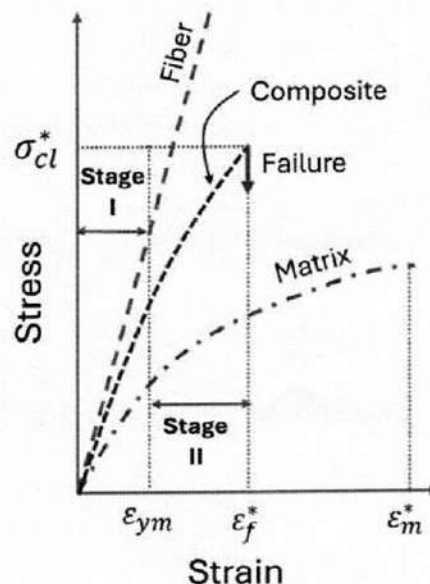


Fig. 5(c): Stress-strain behaviour of the fiber, matrix and composite.

- Q6. (a) Write a short note on the epoxy resin matrix and its chemical formula. Describe with chemical reaction how the epoxy resin is cured using diamine agents. (10)
- (b) What is a coupling agent? Explain how the silane coupling agent can bond with epoxy matrix. (10)
- (c) A composite consists of 30 vol% of glass fibers having a modulus of elasticity 80 GPa and 70 vol% of polyester resin with modulus of 10 GPa. The stress is applied in the longitudinal direction. (15)
- (i) Compute the modulus of elasticity of this composite in the longitudinal direction.
- (ii) If the cross-sectional area is  $300 \text{ mm}^2$  and a stress of 75 MPa is applied in this longitudinal direction, compute the magnitude of the load carried by the fiber and matrix.

- Q7. (a) Briefly describe any two PMC fabrication techniques from below: (10)
- Pultrusion
  - Resin transfer molding
  - Prepreg production.
- (b) Explain the adhesive and cohesive failure of composites with neat sketches. (10)
- (c) Suppose you are a Scientist at Bangladesh Atomic Energy Commission. You are developing a new kind of jute-epoxy multilayer hybrid composite, for which the measured stress-strain curves are shown in Fig. 7(c). (15)
- For the case of unidirectional jute fiber, can you explain why does the stress-strain line change, when fiber orientation angle  $\theta$  is increased from  $0^\circ$  to  $45^\circ$  to  $90^\circ$ ?
  - In case of the bidirectional jute fibers, draw the stress-strain lines for  $\theta = 0^\circ, 45^\circ$  and  $90^\circ$ . Explain why they are different than the unidirectional case.
  - Draw the stress-strain lines for random oriented fiber, when the angle is changed by  $\theta = 0^\circ, 45^\circ$  and  $90^\circ$ . Explain why they are different than both the unidirectional and bidirectional case.

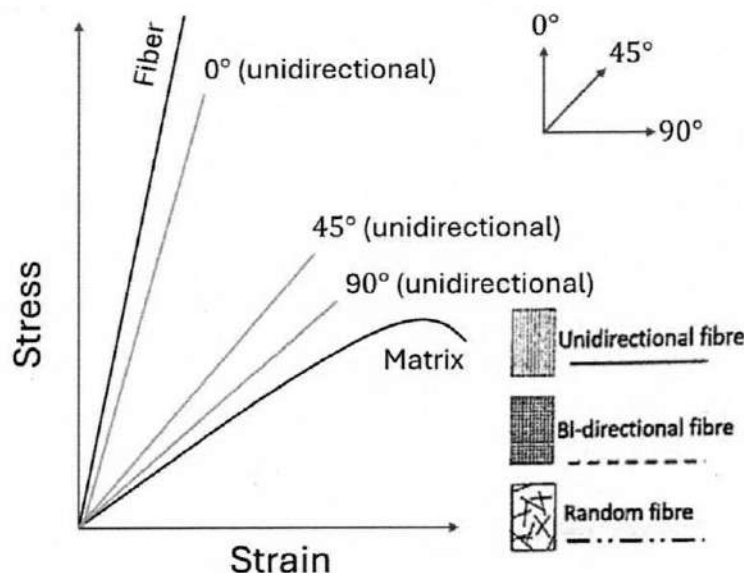


Fig. 7(c): Stress-strain response of the fiber, matrix and composite.

- Q8. (a) Describe any two non-destructive testing methods that can detect cracks deep inside a composite material. (10)
- (b) Write down the advantages and disadvantages of in-situ fabrication of MMC. (10)
- (c) Suppose you are a lead engineer in a ceramic manufacturing company. The company wants your help in producing ceramic matrix composites (CMC) with SiC matrix and SiC fibers. They currently have a large stock of preceramic polymer 'polycarbosilanes' which they want to use. (15)
- Justify which fabrication method you will choose. With neat sketches, explain the manufacturing steps of that method.
  - The company has a future plan to stop using any preceramic polymer. They also don't want to use any kind of liquid infiltration. Which method will you suggest to them for future production? Explain the details of your chosen method with diagram. Also, point out two benefits for the new process.



Khulna University of Engineering & Technology  
B. Sc. Engineering 4<sup>th</sup> Year 1<sup>st</sup> Term Regular Examination, 2023  
Department of Materials Science and Engineering  
MSE 4107

Computational Materials Science

Time: 3 hours

Full Marks: 210

- N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figures in the right margin indicate full marks.

**Section A**

- Q1. (a) Hartree method is one of the early first-principles calculations methods which had certain limitations. Describe Hartree method with its limitations. Also explain suitable remedies for the limitations. (15)
- (b) Discuss Born-Oppenheimer Approximation mentioning the mathematical expression and a suitable benefit. (10)
- (c) Radial electron density represents the arrangement of electrons in core and valence shells. Explain the differences in electron densities of  $Mg^+$  ion and Na atom with neat sketches. (10)
- Q2. (a) Describe the Kohn-Sham approach with suitable figures and also explain how the Kohn-Sham approach convert the interacting Hartree Fock system into non-interacting one electron system. (12)
- (b) Exchange-correlation energy plays a vital role in determining the accuracy of a certain method in first-principles calculations. Discuss exchange-correlation energy briefly and explain the importance of it with reference to Figure Q2(b). (13)

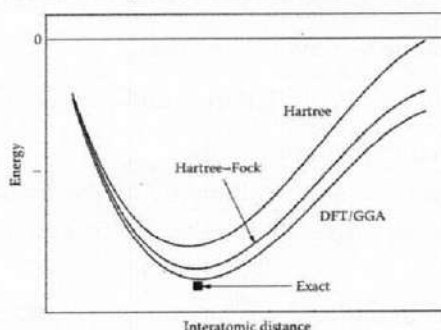


Figure for Q2(b)

- (c) Suggest a suitable XC functional for calculation of lattice parameter of Aluminium, with proper reasoning. Also mention any limitations of the chosen XC functional. (10)
- Q3. (a) Discuss the self-consistent procedure to solve Kohn-Sham equations with appropriate explanation and flow charts. (15)
- (b) Write a short note on “Spin-polarized DFT” and “Time-dependent DFT”. (08)
- (c) Figures Q3(c) (i) and (ii) show the results of cellular automata using two distinct Rule. Identify the Rules for the Figure Q3(c) and differentiate between the Rules, using the Figures Q3(c) (i) and (ii). (12)

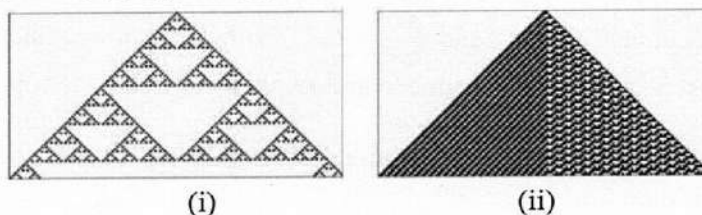


Figure for Q3(c)

- Q4. (a) Which local environment is used in cellular automata for “The Game of Life” prediction? Using reference to the local environment, discuss about “The Game of Life” mentioning the rules used. (12)
- (b) Explain Hesselbarth and Gobel’s model of cellular automata for the recrystallization of materials. (12)
- (c) Draw schematic diagrams to illustrate the following local environment (06)
- (i) Von Neumann environment (ii) Moore environment
- (d) List out the applications of cellular automata. (05)

### Section B

- Q5. (a) Choose a suitable potential for metals & covalent solid and describe the potential with schematic diagram. (12)
- (b) The general expression of Lennard-Jones potential is  $\varphi(r) = 4\epsilon \left[ \left( \frac{\sigma}{r} \right)^{12} - \left( \frac{\sigma}{r} \right)^6 \right]$ , where the parameters have usual meanings. Now determine Lennard-Jones force. (08)
- (c) The general expression of Mie potential is  $\varphi_{mn}(r) = \frac{\epsilon}{m-n} \left( \frac{m}{n} \right)^{\frac{1}{m-n}} \left( \left( \frac{\sigma}{r} \right)^m - \left( \frac{\sigma}{r} \right)^n \right)$ , where the parameters have usual meanings. Now prove that minimum energy is located at an interatomic distance,  $r_m = \left( \frac{m}{n} \right)^{\frac{1}{m-n}} \sigma$ . Also, prove that Mie potential is an equivalent form to the Lennard-Jones potential. (15)

- Q6. (a) Explain the basic principles of the velocity Verlet algorithm. (12)
- (b) Consider a one-dimensional harmonic oscillator of mass,  $m$ , that follows Hooke’s law,  $F = -kx$ , in an isolated system ( $E = \text{constant}$ ). Then Newton’s second law becomes the second-order ordinary differential equation, (15)

$$a(t) = -\frac{k}{m}x(t).$$

Let us assume that the spring is fixed at one side and is an ideal spring with zero mass and spring constant  $k$ . Given the oscillator frequency  $\omega = \sqrt{\frac{k}{m}} = 2$  and initial conditions  $x_0 = 4$  and  $v_0 = 0$ , derive and find the followings:

- (i) Derive the potential expression for the system.
- (ii) Numerical solution for ball positions with time by taking the first two terms in the Taylor series expansions (the Euler method) with  $\Delta t = 0.2$  for 4 timesteps.
- (c) What is potential cutoff in molecular dynamics simulation? Explain why potential cutoff is necessary in molecular dynamics simulation. (08)
- Q7. (a) Explain how thermodynamics properties and dynamical properties are calculated from the molecular dynamics. (15)
- (b) In a simulation in the microcanonical ensemble, the temperature of a system is to be kept constant, to determine specific properties. Explain how the temperature can be kept constant with necessary figure. (12)
- (c) Discuss how molecular dynamics find minimum energy using simulated annealing. (08)
- Q8. (a) How does time average differ from ensemble average? (06)
- (b) Consider the simple two-state system in which energy in state 1 is  $E_1$  and in state 2 is  $E_2$ . Assume  $E_2 > E_1$ , and  $\frac{\Delta E}{k_B T} = 0.8$ , with the starting state as state 1. Generate a series of random number and explain how the Metropolis algorithm works. (15)
- (c) How does the Ising Model determine magnetization in materials using the Metropolis Monte Carlo method? (14)

Khulna University of Engineering & Technology  
B. Sc. Engineering 4<sup>th</sup> Year 1<sup>st</sup> Term Regular Examination, 2023  
Department of Materials Science and Engineering  
MSE 4109

Electronic, Magnetic and Optical Properties of Materials

Time: 3 hours

Full Marks: 210

- N.B.: (i) Answer **ANY THREE** questions from each section in separate scripts.  
(ii) Figures in the right margin indicate full marks.

**Section A**

- Q1. (a) What is the Materials Scientist version of Ohm's law? What are the advantages of this version of Ohm's law? (06)
- (b) Suppose you have to select a copper alloy for the windings of an electromagnet designed to create very high magnetic field. The windings must sustain stresses much higher than pure copper can handle but alloying copper to increase its strength usually also decrease its conductivity which leads to undesirable increased heating of the windings. To get an increase in strength with minimum decrease in conductivity, should you use soluble or insoluble alloying elements? Justify your answer with proper examples. (09)
- (c) Derive an equation that relates microscopic electronic polarization ( $\alpha_e$ ) to the macroscopic relative permittivity ( $\epsilon_r$ ). (08)
- (d) Because of a shortage of Cu during World War II, the electromagnets used in Oak Ridge to separate U-235 isotopes for the Hiroshima atomic bomb were wound with pure silver, which has an atomic volume of 10.30 cc/mol. If 100 m of silver wire 2 mm in diameter required 8.1 watts electrical power when carrying 4 amps of current, calculate (i) resistivity, (ii) electron mobility, (iii) collision time, (iv) mean free path, (v) Hall coefficient, and (vi) Hall field in a transverse magnetic field of 12 tesla. Assume  $T = 325$  K, silver is monovalent and any other missing data. (12)
- Q2. (a) According to Debye's model 'heating a cup of tea in a microwave oven is due to orientation polarization' – explain in brief. (12)
- (b) The electronic polarizability of argon gas is  $1.74 \times 10^{-40}$  farad.m<sup>2</sup>. Calculate the dielectric susceptibility and dielectric constant of argon gas under standard temperature and pressure. Assume an inert gas, which has  $2.69 \times 10^{25}$  molecules per m<sup>3</sup> at standard temperature and pressure. (08)
- (c) Illustrate the physical significance of wave function. Explain the wave-particle duality of electrons with necessary figures by a combination of several wave trains having slightly different frequencies. (15)
- Q3. (a) What is the Heisenberg uncertainty principle? Explain this principle by applying Schrodinger's wave equation in the case of an electron motion in free space. (11)
- (b) Find out the electron energy,  $E = 0.12$  eV tunnelling through a potential barrier of height  $V_0 = 1.2$  eV and width = 5 Å. (08)
- (c) For a free electron, calculate the density of states per unit volume (cm<sup>3</sup>) over the energy range of (i)  $0 \leq E \leq 2$  eV and (ii)  $1 \leq E \leq 2$  eV. (08)
- (d) Show that the maximum in the radial probability function for the 1s electron state in hydrogen occurs at  $r = a_0$ . Also find the value of  $a_0$ . (08)

- Q4. (a) Demonstrate the splitting of outer shell electronic states of Si atoms as a function of interatomic distance with a schematic diagram. State the possible number of occupied and empty states before and after splitting assuming that there is only one gram-mole of Si with periodic atomic arrangement. (10)
- (b) Demonstrate that the first derivative of  $E$  with respect to  $k$  is related to the velocity and the second derivative is related to the mass of a free electron. Use these results to explain the concept of effective mass and prove that the effective mass of an electron near the top of the valance band is negative. (12)
- (c) Determine the relative effective mass ( $m^*/m_0$ ) of conduction band electrons within a crystal lattice having a lattice parameter  $a = 10 \text{ \AA}$ . Assume that the electronic state of the conduction band follows a parabolic  $E$  vs.  $k$  relationship; where at  $k = \pm \pi/a$ ,  $E = E_c + 0.32 \text{ eV}$  and at  $k = 0$ ,  $E = E_c$ . (05)
- (d) Calculate the number of quantum states per  $\text{cm}^3$  in silicon between  $E_c$  and  $(E_c + 0.35) \text{ eV}$  at  $T = 300 \text{ K}$  using the basic principle of infinite potential well. (08)

### Section B

- Q5. (a) Briefly explain why all materials are not magnets. (08)
- (b) 'Magnetic property in a material depends on its electronic configuration' – Justify the statement using at least two examples. (12)
- (c) Consider a ferromagnetic particle consisting of  $10^6$  atoms, each of which has a magnetic dipole moment of two Bohr magnetons. In an H-field of  $10^6 \text{ A/m}$ , if the total magnetic dipole moment of the particle rotates from antiparallel to the field to parallel to the field, by how much does its energy decrease? For what temperature will this equal  $k_B T$ ? (15)
- Q6. (a) Discuss the magnetization curve of a soft magnetic material having impurities, structural defects such as particles of a nonmagnetic phase or voids. (10)
- (b) Describe with illustration, the BCS theory of superconductivity in terms of electro-phonon interaction. (15)
- (c) Describe the theory which supports that electrical conductivity in a material can be increased to infinity. (10)
- Q7. (a) Effect of inhomogeneities or defects enables a material (type-II superconductor) to gain hard ferromagnet-like hysteresis behaviour – explain the statement with necessary figure. (15)
- (b) The fraction of nonreflected light that is transmitted through a 200 mm thickness of glass is 0.98. Calculate the absorption coefficient of this material. (10)
- (c) "Radiation is amplified in LASER" – Justify the statement with diagram. (10)
- Q8. (a) A beam of red light with a wavelength in vacuum of 700 nm enters an insulator with an effective dielectric constant  $\epsilon_r$  of 1.7 at the frequencies of visible light. What will be the frequency, angular frequency, velocity, and wavelength of this light when travelling within the insulator? (12)
- (b) Write short notes on – (i) Rayleigh scattering, (ii) Compton scattering, (iii) Photoelectric effect, and (iv) Tyndall effect. (08)
- (c) 'A good conductor such as Cu is a good reflector in the IR region' – Prove it with mathematical expression. (15)